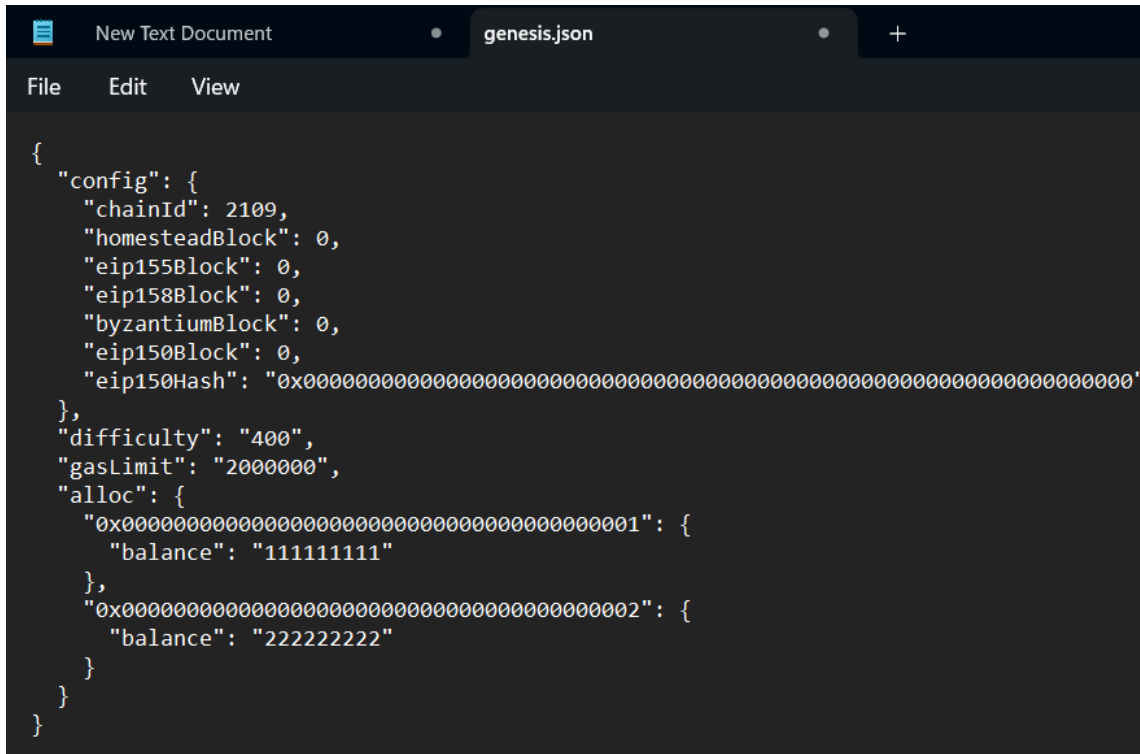


42000 Cryptography Implementation Project Ethereum

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Task: Run your private Ethereum network with a single node.

A screenshot of a text editor window with a dark theme. The window has two tabs: 'New Text Document' and 'genesis.json'. The 'genesis.json' tab is active. The editor shows a JSON configuration for an Ethereum genesis block. The configuration includes a 'config' object with 'chainId', 'homesteadBlock', 'eip155Block', 'eip158Block', 'byzantiumBlock', 'eip150Block', and 'eip150Hash'. It also includes 'difficulty', 'gasLimit', and an 'alloc' object with two initial accounts, each having a 'balance'.

```
{
  "config": {
    "chainId": 2109,
    "homesteadBlock": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "eip150Block": 0,
    "eip150Hash": "0x0000000000000000000000000000000000000000000000000000000000000000"
  },
  "difficulty": "400",
  "gasLimit": "2000000",
  "alloc": {
    "0x0000000000000000000000000000000000000000000000000000000000000001": {
      "balance": "11111111"
    },
    "0x0000000000000000000000000000000000000000000000000000000000000002": {
      "balance": "22222222"
    }
  }
}
```

Fig 1.1 Genesis.json file that provides the software with its initial values.

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.22621.2861]
(c) Microsoft Corporation. All rights reserved.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b>geth init C:\Users\khela\OneDrive\Des
ktop\genesis.json --datadir D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b
INFO [04-05|19:38:35.208] Maximum peer count ETH=50 LES=0 total=50
INFO [04-05|19:38:35.303] Allocated cache and file handles database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata cache=16.00MiB handles=16
INFO [04-05|19:38:35.330] Writing custom genesis block
INFO [04-05|19:38:35.334] Persisted trie from memory database nodes=3 size=399.00B time=0s gcnodes=0 gcsiz=0.00B g
ctime=0s livenodes=1 liveness=0.00B
INFO [04-05|19:38:35.350] Successfully wrote genesis state database=chaindata hash="0dc6c4...45e892"
INFO [04-05|19:38:35.355] Allocated cache and file handles database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\lightchaindata cache=16.00MiB handles=16
INFO [04-05|19:38:35.377] Writing custom genesis block
INFO [04-05|19:38:35.381] Persisted trie from memory database nodes=3 size=399.00B time="991.8µs" gcnodes=0 gcsiz=
0.00B gctime=0s livenodes=1 liveness=0.00B
INFO [04-05|19:38:35.395] Successfully wrote genesis state database=lightchaindata hash="0dc6c4...45e892"

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b>geth init C:\Users\khela\OneDrive\Des
ktop\genesis.json --datadir D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b --network
id 8080
Incorrect Usage: flag provided but not defined: -networkid

init [command options] [arguments...]

The init command initializes a new genesis block and definition for the network.
This is a destructive action and changes the network in which you will be
participating.
```

Fig1.2 Initialising the Ethereum network

```
C:\Windows\System32\cmd.e x + v
D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b>geth --datadir D:\Downloads\geth-wind
ows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b --networkid 8080
INFO [04-05|19:44:14.150] Maximum peer count ETH=50 LES=0 total=50
INFO [04-05|19:44:14.237] Starting peer-to-peer node instance=Geth/v1.9.15-stable-0f77f34b/windows-amd64/g
o1.14.2
INFO [04-05|19:44:14.246] Allocated trie memory caches clean=256.00MiB dirty=256.00MiB
INFO [04-05|19:44:14.250] Allocated cache and file handles database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata cache=512.00MiB handles=8192
INFO [04-05|19:44:14.305] Opened ancient database database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata\ancient
INFO [04-05|19:44:14.315] Initialised chain configuration config="{ChainID: 2109 Homestead: 0 DAO: <nil> DAOSup
port: false EIP150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: <nil> Petersburg: <nil> Istanbul: <nil>, Muir Gla
cier: <nil>, YOLO v1: <nil>, Engine: unknown}"
INFO [04-05|19:44:14.327] Disk storage enabled for ethash caches dir=D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\g
eth-windows-amd64-1.9.15-0f77f34b\geth\ethash count=3
INFO [04-05|19:44:14.334] Disk storage enabled for ethash DAGs dir=C:\Users\khela\AppData\Local\Ethash count=2
INFO [04-05|19:44:14.339] Initialising Ethereum protocol versions="[65 64 63]" network=8080 dbversion=<nil>
WARN [04-05|19:44:14.344] Upgrade blockchain database version from=<nil> to=7
INFO [04-05|19:44:14.347] Loaded most recent local header number=0 hash="0dc6c4...45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.351] Loaded most recent local full block number=0 hash="0dc6c4...45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.354] Loaded most recent local fast block number=0 hash="0dc6c4...45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.360] Regenerated local transaction journal transactions=0 accounts=0
INFO [04-05|19:44:14.372] Allocated fast sync bloom size=512.00MiB
INFO [04-05|19:44:14.377] Initialized fast sync bloom items=3 errorrate=0.000 elapsed=0s
INFO [04-05|19:44:14.399] New local node record seq=1 id=6e08d042bb9efea ip=127.0.0.1 udp=30303 tcp=
30303
INFO [04-05|19:44:14.405] Started P2P networking self=enode://4f577e0e311a46849f9fa27a1ba7b231c1c336a4
ed4b0932c527f4a69d30c138daf887625caf902be7666ef072e8c0f58bf2c3325dadbd2d04766d76499e74d6c@127.0.0.1:30303
INFO [04-05|19:44:14.405] IPC endpoint opened url=\\.\pipe\geth.ipc
```

Fig 1.3 Running the network with the console enabled.

```
C:\Windows\System32\cmd.e x + v
INFO [04-05|19:44:14.377] Initialized fast sync bloom
INFO [04-05|19:44:14.399] New local node record
30303
INFO [04-05|19:44:14.405] Started P2P networking
ed4b0932c527f4a69d30c138daf887625caf902be7666ef072e8c0f58bf2c3325dadb2d04766d76499e74d6c@127.0.0.1:30303
INFO [04-05|19:44:14.405] IPC endpoint opened
INFO [04-05|19:44:22.167] New local node record
p=30303
INFO [04-05|19:44:24.416] Looking for peers
INFO [04-05|19:44:36.320] Looking for peers
INFO [04-05|19:44:46.722] Looking for peers
INFO [04-05|19:44:56.723] Looking for peers
INFO [04-05|19:45:07.129] Looking for peers
INFO [04-05|19:45:17.221] Looking for peers
INFO [04-05|19:45:27.246] Looking for peers
INFO [04-05|19:45:37.837] Looking for peers
INFO [04-05|19:45:47.862] Looking for peers
INFO [04-05|19:45:57.868] Looking for peers
WARN [04-05|19:46:02.233] Served eth_coinbase
ied"
INFO [04-05|19:46:08.045] Looking for peers
INFO [04-05|19:46:18.097] Looking for peers
INFO [04-05|19:46:28.136] Looking for peers
INFO [04-05|19:46:38.249] Looking for peers
INFO [04-05|19:46:48.297] Looking for peers
INFO [04-05|19:46:58.312] Looking for peers
INFO [04-05|19:47:08.357] Looking for peers
INFO [04-05|19:47:18.498] Looking for peers
INFO [04-05|19:47:28.532] Looking for peers
INFO [04-05|19:47:38.555] Looking for peers

items=3 errorrate=0.000 elapsed=0s
seq=1 id=6e08d042bb9efee ip=127.0.0.1 udp=30303 tcp=
self=enode://4f577e0e311a46849f9fa27a1ba7b231c1c336a4
url=\\.pipe\geth.ipc
seq=2 id=6e08d042bb9efee ip=138.25.4.65 udp=30303 tc
peercount=0 tried=106 static=0
peercount=0 tried=101 static=0
peercount=0 tried=107 static=0
peercount=0 tried=134 static=0
peercount=0 tried=90 static=0
peercount=0 tried=122 static=0
peercount=0 tried=59 static=0
peercount=0 tried=126 static=0
peercount=0 tried=179 static=0
peercount=0 tried=144 static=0
reqid=3 t=0s err="etherbase must be explicitly specif
```

Fig 1.4 Playing with some supported commands.

Task 2: Create accounts and Mining for tokens.

g

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.22621.2861]
(c) Microsoft Corporation. All rights reserved.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>attach \\.pipe\geth.ipc
'attach' is not recognized as an internal or external command,
operable program or batch file.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth attach \\.pipe\geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.9.15-stable-0f77f34b/windows-amd64/go1.14.2
at block: 0 (Thu Jan 01 1970 11:00:00 GMT+1100 (AEDT))
datadir: D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b
modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

> personal.newAccount()
Passphrase:
Repeat passphrase:
"0x3cca35d35f682508499743b20c2f61a7ae72b45b"
> personal.newAccount()
Passphrase:
Repeat passphrase:
"0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"
> |
```

Fig 2.1 Creating two accounts.

```
C:\Windows\System32\cmd.e  X + v
D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>attach \\.pipe\geth.ipc
'attach' is not recognized as an internal or external command,
operable program or batch file.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth attach \\.pipe\geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.9.15-stable-0f77f34b/windows-amd64/go1.14.2
at block: 0 (Thu Jan 01 1970 11:00:00 GMT+1100 (AEDT))
datadir: D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b
modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

> personal.newAccount()
Passphrase:
Repeat passphrase:
"0x3cca35d35f682508499743b20c2f61a7ae72b45b"
> personal.newAccount()
Passphrase:
Repeat passphrase:
"0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
>
> miner.setEtherbase("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
true
> eth.coinbase
"0x3cca35d35f682508499743b20c2f61a7ae72b45b"
> |
```

Fig 2.2 Checking the initial values of both the accounts.

Task 3: Create transaction and mine into blocks

```
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> eth.accounts
["0x3cca35d35f682508499743b20c2f61a7ae72b45b", "0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"]
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> miner.start(1)
null
> miner.stop()
null
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
9000000000000000000
> eth.blockNumber
3
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
9
> web3.fromWei(eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"), "ether")
0
> |
```

Fig 3.1 Balance of both the accounts after mining.


```
C:\Windows\System32\cmd.e  X  +  v

["0x3cca35d35f682508499743b20c2f61a7ae72b45b", "0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"]
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> miner.start(1)
null
> miner.stop()
null
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
9000000000000000000
> eth.blockNumber
3
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
9
> web3.fromWei(eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"), "ether")
0
> eth.getBalance(eth.accounts[0])
9000000000000000000
> personal.unlockAccount(eth.accounts[0], "1111")
true
> eth.sendTransaction({from:eth.accounts[0], to:eth.accounts[1], value:web3.toWei(50, "ether")})
Error: insufficient funds for transfer
    at web3.js:6347:37(47)
    at web3.js:5081:62(37)
    at <eval>:1:20(21)

> eth.sendTransaction({from:eth.accounts[0], to:eth.accounts[1], value:web3.toWei(2, "ether")})
"0x222a9a3412f7e28ed8827dc9396ab72dd5732d3dd80f8f7789ed33818b91308e"
> |
```

Fig 3.2 Transaction from base account to the second account

```
C:\Windows\System32\cmd.e X + v
> eth.sendTransaction({from:eth.accounts[0], to:eth.accounts[1], value:web3.toWei(2, "ether")}]
"0x222a9a3412f7e28ed8827dc9396ab72dd5732d3dd80f8f7789ed33818b91308e"
> miner.start(1)
null
> miner.stop()
null
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
13
> web3.fromWei(eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"), "ether")
2
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
13
> eth.getTransaction("0x222a9a3412f7e28ed8827dc9396ab72dd5732d3dd80f8f7789ed33818b91308e")
{
  blockHash: "0x87a0b7cafaa9a72e84bef68538acf0444af2ca8667d6b89b72042164aa828408",
  blockNumber: 4,
  from: "0x3cca35d35f682508499743b20c2f61a7ae72b45b",
  gas: 21000,
  gasPrice: 1000000000,
  hash: "0x222a9a3412f7e28ed8827dc9396ab72dd5732d3dd80f8f7789ed33818b91308e",
  input: "0x",
  nonce: 0,
  r: "0xd207b41465805e5c0ae95f3b3fabe003eb11b02419ce3942c2dd688e574e38ac",
  s: "0xb369c99713d60688b0e3fb8466dbeb4484f549e68235128466724d4030e011d",
  to: "0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d",
  transactionIndex: 0,
  v: "0x109e",
  value: 2000000000000000000
}
> |
```

Fig 3.3 Balance of the two accounts and retrieval of transaction with its hash value.

To get things going, we made two accounts: one called the "etherbase" account to enable transactions. We used "miner.setEthbase" to start mining, and then we used "eth.sendTransaction" to start a transaction between the two accounts. The analysis of the transaction is as follows

- ❑ The hash value of block number 4, where the transaction occurred, is known as the blockhash.
- ❑ From indicates the ethbase account
- ❑ The quantity of 21,000 wei was the transaction gas, which denotes the expense of executing a function or transaction on the Ethereum network.
- ❑ The "gasPrice" is set at 1,000,000,000 wei, which is equivalent to 1 gwei.
- ❑ Hash value indicates the hash of the transaction.
- ❑ The input is denoted by "0x" since it refers to the data that was transmitted along with the transaction, which in our case was none.
- ❑ Since this is the first transaction, there have never been any others, which explains why the nonce is 0.

- ☐ The r and s values are typical outputs of a standard Elliptic Curve Digital Signature Algorithm (ECDSA) signature.
- ☐ The recipient account, in this case our second account, is specified via the "To" field.
- ☐ Since this is the first transaction, its transaction index is 0
- ☐ V is needed to recover the public key.
- ☐ The value denotes the amount in wei that was transferred.